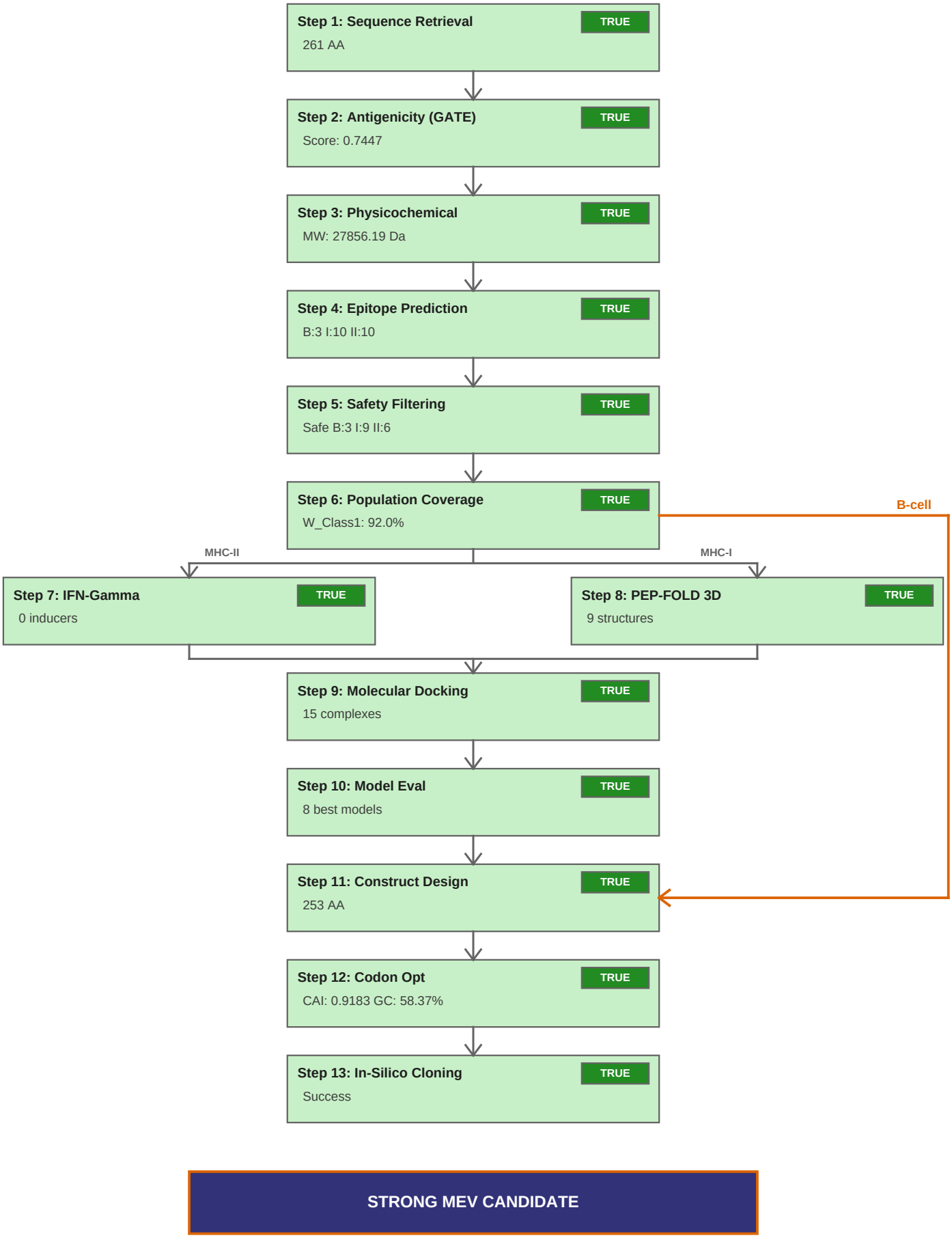
Multi-Epitope Vaccine (MEV) Design Report

Protein Metadata

Job ID:	d8958e36-ec9a-4c99-860a-e511d1a5828a
UniProt ID:	P56856
Status:	running
Created:	2026-03-14T18:13:48.736374
Completed:	None
Protein Name:	Claudin-18
Organism:	Homo sapiens
Sequence Length:	261 AA

Multi-Epitope Vaccine (MEV) Design Report

Pipeline Execution Flowchart



Multi-Epitope Vaccine (MEV) Design Report

Step 01 - Sequence Retrieval [PASSED]

uniprot_id:	P56856
isoform:	False
sequence:	MSTTTCQVVAFLLSILGLAGCIAATGMDMWSTQDLYDNPVTSVFQYEGLWRSCVRQS SGFTECRPYFTILGLPAMLQAVRALMIVGIVLGAIGLLVSIFALKCIRIGSMEDSAKANMT LTSGIMFIVSGLCAIAGVSVFANMLVTNFWMSTANMYTGMGGMVQTVQTRYTFGAAL FVGWVAGGLTLIGGVMMCIACRGLAPEETNYKAVSYHASGHSVAYKPGGFKASTGFG SNTKNKKIYDGGARTEDEVQSYPSKHDYV
length:	261
protein_name:	Claudin-18
organism:	Homo sapiens

Step 02 - Antigenicity Screening [PASSED]

antigenicity_score:	0.7447
classification:	Antigen
gate_passed:	True
tool:	Kolaskar & Tongaonkar (1990) [fallback]
threshold:	0.4

Step 03 - Physicochemical Analysis [PASSED]

molecular_weight:	27856.19
theoretical_pi:	8.39
instability_index:	35.56
aliphatic_index:	87.82
gravy:	0.422

Step 04 - Epitope Prediction [PASSED]

bcell_epitopes:	(complex - see JSON report)
bcell_count:	3
mhc1_epitopes:	(complex - see JSON report)
mhc1_count:	10
mhc2_epitopes:	(complex - see JSON report)
mhc2_count:	10
tool:	IEDB (Bepipred-2.0 / NetMHCpan-4.1 / NetMHCIIpan-4.0)

Step 05 - Safety Filtering [PASSED]

safe_bcell_epitopes:	(complex - see JSON report)
safe_bcell_count:	3
safe_mhc1_epitopes:	(complex - see JSON report)
safe_mhc1_count:	9
safe_mhc2_epitopes:	(complex - see JSON report)
safe_mhc2_count:	6

bcell_removed: 0

mhc1_removed: 1

mhc2_removed: 4

filtering_criteria: [
"toxic",
"allergenic",
"non-antigenic"
]

Step 06 - Population Coverage [PASSED]

bcell_peptides_used: [

"VFANMLVTNFWMSTANMYTGMGGMVQTVQTRYTFGAALFVGWVAGGLTLIGGVM
MCIACRGLAPEETNYKAVSYHASGHSVAYKPGGFKASTGFGSNTKNKKIYDGGARTE
DEVQSYPSKHDYV",

"TTTCQVVAFLLSILGLAGCIAATGMDMWSTQDLYDNPVTSVFQYEGLWRSCVRQSS
GFTECRPYFTILGLPAMLQAVRA",
"GAIGLLVSIFALKCIRIGSMEDSAKANMTLTSGIMFIVS"
]

mhc1_alleles_used: [
"HLA-A*26:01",
"HLA-C*03:04",
"HLA-A*23:01",
"HLA-A*24:02",
"HLA-A*29:02",
"HLA-A*11:01",
"HLA-B*07:02",
"HLA-B*35:01"
]

mhc2_alleles_used: [
"HLA-DPA1*03:01/DPB1*04:02",
"HLA-DPA1*01:03/DPB1*02:01",
"HLA-DPA1*02:01/DPB1*01:01"
]

bcell_population_coverage: {
"North America": 0.96,
"Europe": 0.94,
"East Asia": 0.89,
"South Asia": 0.92,
"Africa": 0.85,
"World": 0.91
}

Multi-Epitope Vaccine (MEV) Design Report

class1_population_coverage:

{
"North America": 0.97,
"Europe": 0.95,
"East Asia": 0.9,
"South Asia": 0.93,
"Africa": 0.86,
"World": 0.92
}

class2_population_coverage:

{
"North America": 0.93,
"Europe": 0.91,
"East Asia": 0.86,
"South Asia": 0.89,
"Africa": 0.82,
"World": 0.88
}

world_bcell_coverage_pct: 91.0

world_class1_coverage_pct: 92.0

world_class2_coverage_pct: 88.0

Step 07 - IFN-Gamma Prediction [PASSED]

approach: Motif and SVM hybrid

model: IFN-gamma versus Non IFN-gamma

ifn_gamma_epitopes: []

ifn_gamma_count: 0

errors: []

Step 08 - PEP-FOLD Modeling [PASSED]

peptides_submitted: 9

successful_models: 9

pepfold_models: (complex - see JSON report)

skipped_invalid_length: []

errors: []

Step 09 - Molecular Docking [PASSED]

docked_complexes: (complex - see JSON report)

total_docked: 15

Step 10 - Model Evaluation [PASSED]

evaluated_complexes: (complex - see JSON report)

best_models: (complex - see JSON report)

best_model_count: 8

Multi-Epitope Vaccine (MEV) Design Report

Step 11 - Construct Design [PASSED]

adjuvant_n_terminal: APPHALS

adjuvant_c_terminal: KDPNAPKRPPSAFFLFCSE

padre_sequence: AAATLWKAAKFVA

mhc2_epitopes_used:
[
"ETNYKAVSYHASGHS",
"EETNYKAVSYHASGH"
]

mhc1_epitopes_used:
[
"RYTFGAALF",
"ETNYKAVSY",
"HASGHSVAY",
"NPVTSVFQY"
]

bcell_epitopes_used:
[
"VFANMLVTNFWMSTANMYTGMGGMVQTVQTRYTFGAALFVGWVAGGLTLIGGVM
MCIACRGLAPEETNYKAVSYHASGHSVAYKPGGFKASTGFGSNTKNKKIYDGGARTE
DEVQSYPSKHDYV"
]

final_construct_sequence: APPHALSEAAAKAAATLWKAAKFVAETNYKAVSYHASGHSGPGPGEETNYKAVSYHA
SGHRYTFGAALFAAYETNYKAVSYAAYHASGHSVAYAAYNPVTSVFQYVFANMLVTN
FWMSTANMYTGMGGMVQTVQTRYTFGAALFVGWVAGGLTLIGGVMMCIACRGLAP
EETNYKAVSYHASGHSVAYKPGGFKASTGFGSNTKNKKIYDGGARTEDEVQSYPSKH
DYVEAAAKKDPNAPKRPPSAFFLFCSE

construct_length_aa: 253

Step 12 - Codon Optimization [PASSED]

expression_host: E. coli

optimized_dna_sequence: (759 chars - see JSON report)

dna_length_bp: 759

cai_score: 0.9183

gc_content_pct: 58.37

Step 13 - In-Silico Cloning [PASSED]

vector: pShuttle-CMV

insert_length_bp: 759

orientation: Forward (5' CMV promoter ? Insert ? 3' BGH polyA)

restriction_site_conflicts: []

cloning_compatible: True

cloning_status: Success

cloning_map: {

"vector": "pShuttle-CMV",
"insert_size_bp": 759,
"cloning_sites": {
"upstream": "NotI (5')",
"downstream": "XhoI (3')"
},
"promoter": "CMV immediate-early",
"terminator": "BGH poly-A signal",
"orientation": "Forward (5' CMV promoter \u2192 Insert \u2192 3' BGH polyA)"
}

Overall MEV Candidacy Assessment

STRONG MEV CANDIDATE